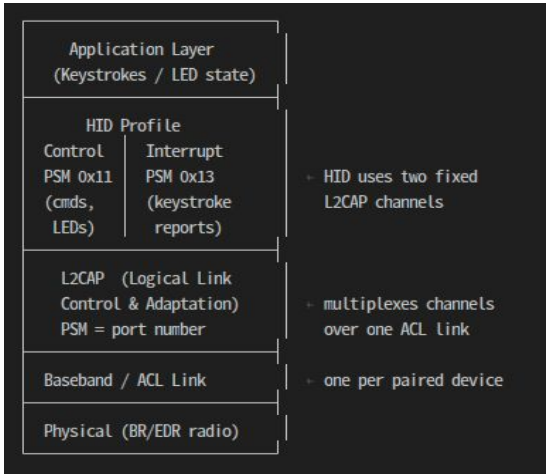
NiNo: How NoInputNoOutput Breaks Bluetooth Pairing Security

Threat Model Assumption: The victim keyboard and PC have never previously paired. Since IO capability exchange only occurs when two victim devices are previously unknown to each other, this assumption is essential for NiNo attack.

Bluetooth Protocol Stack (HID Keyboard)



- SDP (Service Discovery Protocol) runs before connection — the PC queries the keyboard's service record to learn which device to connect to. The attacker must register a matching HID SDP record to connect with the victim PC.

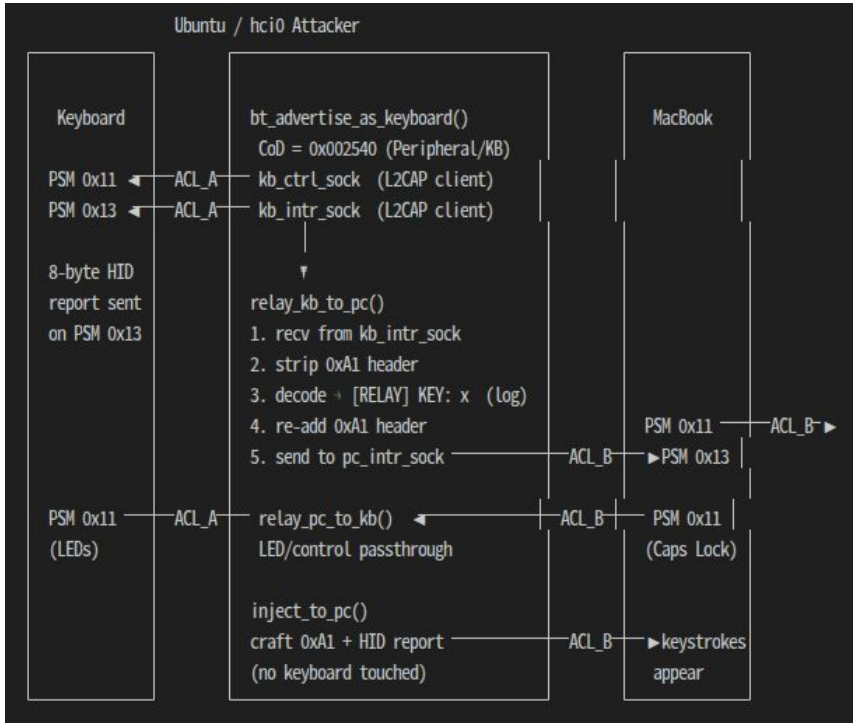
Secure Simple Pairing (SSP) IO Capability Negotiation

Initiator ↓ / Responder →	DisplayOnly	DisplayYesNo	KeyboardOnly	NoInputNoOutput
DisplayOnly	Passkey	Passkey	Passkey	Just Works
DisplayYesNo	Passkey	Numeric Comparison	Passkey	Just Works
KeyboardOnly	Passkey	Passkey	Passkey	Just Works
NoInputNoOutput	Just Works	Just Works	Just Works	Just Works

- If either side claims NoInputNoOutput, the spec mandates Just Works.
- The attacker sets **btmgmt io-cap 3** on hci0 to claim this capability before pairing with both the keyboard and the PC. Neither side can object.

Implementation: Single-Adapter MITM Relay Using NiNo

Architecture Diagram



1. SDP Registration

- The attacker calls **sdptool add HID** to register a HID service record.
- The SDP record tells the target PC the HID PSM numbers and device class.

2. The 0xA1 Header

- Every HID interrupt packet from a real keyboard begins with 0xA1 (HID transaction type: DATA | INPUT).
- The MITM must strip this byte before decoding the 8-byte report, and re-add it before forwarding to victim PC.

3. Scatternet Feasibility

- A single hci0 holds two ACL connections simultaneously (BlueZ scatternet).
- No BD_ADDR spoofing is needed for NiNo — the PC has never seen the keyboard before (initial pairing), so it pairs with whatever device advertises as a keyboard.
- The attacker wins the pairing race in this demo by disabling PC Bluetooth during the attack window, simulating the presence of a jammer.